

Deadline for submission: 20th May, 2026

Deliverables: The project will be reported in writing and presentation. The report is expected to give a full account of your solutions and results, and it should follow the IMRAD-structure of an **academic report**. There is a page limit of max. 4 pages, including figures and references and assuming a font size of 10pt. In addition, submitting your solution in the form of **Python code**.

Background

- **Goal:** Predict the moisture content of wood using sensor measurements.
- Traditionally, moisture content has been determined through oven drying and weighting (Destructive approach)
- **Why it matters:**
 - Reduces carbon emissions (less oven drying).
 - Saves material (non-destructive method).
 - Helps maintain optimal calorific value (efficient burning).



Figure: Data acquisition system

Understand the Data

You have a dataset

where:

Columns 1–6: Sensor measurements (features / inputs). (see the above data acquisition setup where sensors are placed to make the measurements)

Last column: Moisture content (target / output).

Problem Statement

The aim of this assignment is to predict the moisture content of wood using sensor measurements. Unlike traditional oven-drying methods (destructive and energy-intensive), this approach uses sensor data to enable faster and more sustainable predictions.

Since the dataset is partial (only 6 sensors), perfect accuracy is not expected. The focus is on applying machine learning correctly and interpreting results

Objective

- Build regression models to predict moisture content.
- Identify **which sensor(s) contribute most** to prediction.
- Evaluate model performance using appropriate metrics.
- Analyze limitations due to incomplete data.

Data Preprocessing

- Standardized features
- Split dataset into:
 - Training set (e.g., 80%)
 - Testing set (20%)

Models to be Used

- Different versions of **Linear Regression (Discussed in Class)**
- **Neural networks (This will be covered in next class)**